Lesson 16: Coding Scientific Programs

1. What is the main benefit of using simulations?

a) Adding unnecessary risks
b) Testing without real world risk
c) Lowering the accuracy of results
d) Using extra time and resources

2. What makes a simulation effective?

a) Accurate model
b) Complex design
c) Flashy graphics
d) Louder sounds

3. Which three components are essential in a simple circuit simulation?

a) Solar, switch, lamp
b) Wire, motor, fan
c) Battery, bulb, wire
d) Capacitor, diode, motor

4. In the Scratch circuit simulator, what happens when you turn on the switch?

a) The program stops
b) The code disappears
c) The background changes
d) The bulb lights up

5. Why is it important for simulations to follow real rules of science?

a) To win games
b) To match real-life results
c) To confuse the user
d) To make computers talk

6. Why do coders use variables in simulations?

a) To label objects
b) To add sound
c) To store changing values
d) To design backgrounds

7. Why does accuracy matter when coding a simulation?

a) Playing simulation games
b) Improves screen visuals
c) Animating virtual characters
d) Gives realistic results

Answers:

1.b	2.a	3. c	4. d	5. b	6. c	7. d

Lesson 17: Using Simulation to Recreate a Famous Event

1. What makes a simulation effective and reliable for recreating an event?

a) Using an oversimplified model
b) Using a realistic and accurate model
c) Using a model with missing details
d) Using outdated or incorrect data

2. Which technology is often linked with virtual simulations of events?

a) Virtual reality
b) Digital painting
c) Word processing
d) Photo editing

3. Why is running a computer simulation safer than testing a real building in a disaster?

a) Random guessing
b) Confuse people
c) Understand events
d) Make movies

4. What natural disaster is often studied using simulations to protect buildings?

a) Thunderstorms
b) Rainstorms
c) Sunny weather
d) Earthquakes

5. What is the famous ship that sank in 1912?

a) The Queen Mary
b) The Bismarck
c) The Titanic
d) The Lusitania

6. Why are simulations important before building ships or planes?

a) To test safety
b) To make the objects look nicer
c) To only use simple designs
d) To avoid coding

7. What must engineers use to make an accurate simulation?

a) Rumors and guesses
b) Known, reliable facts
c) Facts from movies
d) Only new facts

8. What can earthquake simulations show engineers?

a) Building window size
b) Cost of future repairs
c) Structure failure spots
d) Building designs

9. Why is it important for a simulation to follow the real rules of physics?

a) To stop the program from crashing
b) To speed up animation
c) To speed up coding
d) To ensure results are valid

Answers:

1.b	2.a	3.c	4.d	5.c	6.a	7.b	8. c	9.d

Lesson 18: Creating a Safety Simulation

1. For what purpose are crowd simulations mainly used?

a) Ticket counting
b) Event scheduling
c) Route planning
d) Crowd safety

2. What kind of vehicle routes are planned using safety simulations?

a) Garbage trucks
b) Emergency vehicles
c) School buses
d) Mail vans

3. Which environment needs the most crowd safety planning?

a) Small family dinner
b) Quiet library
c) Large public event
d) Sleeping time

4. What do officials plan for a big events using simulation?

a) Planning seating layouts
b) Scheduling staff teams
c) Organizing entry routes
d) Ensuring public safety

5. What is the process of finding and fixing errors in the simulation code called?

a) Debugging
b) Saving
c) Compiling
d) Testing

6. What do you need to define in the simulation to represent people moving?

a) Their favorite color

b) Their movement speed

c) Their music choice

d) Their height

7. To make a safety simulation highly reliable, what must the coded logic be based on?

a) Opinions of friends

b) Personal guesses

c) Real-world physics

d) Random chance

Answers:

1.d	2.b	3.c	4.d	5. a	6.b	7.c

Lesson 19: 3D Printing

1. What does the 3D in 3D printing stand for?

a) Three drawings
b) Three directions
c) Three dimensions
d) Three different

2. What is the first step in 3D printing?

a) Create design
b) Heat plastic
c) Add colors
d) Shape manually

3. How does a 3D printer create objects?

a) Cutting
b) Layering
c) Painting
d) Welding

4. What kind of objects can 3D printers make?

a) Simple plastic models
b) Decorative items
c) Miniature toys
d) Functional parts and prototypes

5. Which material is commonly used in 3D printing?

a) Plastic
b) Cloth
c) Paper
d) Wood

6. What is the soft, thread-like material used by 3D printers called?

a) Plastic wire

b) Filament

c) Wood chip

d) Soft clay

7. Why is 3D printing considered an additive process?

a) It subtracts material

b) It bends metal

c) It builds layer by layer

d) It paints the final object

8. Which feature makes some 3D prints look so real?

a) Bright colors

b) Simple design

c) Small size

d) Realistic look

Answers:

1.c	2.a	3.b	4.d	5.a	6.b	7.c	8.d

Lesson 20: Programming: Displaying Sound Waves

1. How does sound move?

a) Lines

b) Waves

c) Signals

d) Pulses

2. What does not make a sound wave?

a) Singing

b) Clapping

c) Heartbeat

d) Light

3. What role do sound wave simulations play in architectural planning?

a) Predict sound behavior in layouts

b) Choose paint colors

c) Decide building shape

d) Plan landscaping

4. If you speak louder into the microphone, what happens to the sound wave?

a) Waves get smaller

b) Waves disappear

c) Waves get bigger

d) Waves move slower

5. What aspect of sound does sound wave simulations help researchers study?

a) The taste of sound

b) The behavior of sound waves

c) The color of sound

d) The smell of sound

6. What is the main importance of sound visualization technology?

a) To study and analyze sound

b) To measure only the volume of sound

c) To detect sound patterns without analysis

d) To make sound recordings

7. What does the program create on the screen when it detects sound?

a) A flashing light

b) A series of lines

c) A random drawing

d) A wave pattern

Answers:

1.b	2.d	3.a	4.c	5.b	6.a	7.d

Lesson 21: Programming: Controlling a Physical System

1. What device measures a physical property like heat or light?

a) A monitor
b) A sensor
c) A speaker
d) A timer

2. What does the input part of a control system do?

a) Gathers data
b) Fixes errors
c) Changes colors
d) Prints paper

3. What is an example of a control system that most people have in their homes?

a) A hammer
b) A thermostat
c) A pencil
d) A frying pan

4. What is the main purpose of sensors?

a) Detect changes
b) Control motors
c) Store data
d) Reduce power

5. Which sensor is used to detect movement?

a) Thermostat
b) Fan
c) Motion sensor
d) Speaker

6. What property does a thermostat primarily monitor?

a) Air pressure
b) Room color
c) Wind speed
d) Temperature

7. What do you call a system that keeps a room at a certain temperature?

a) A variable system
b) A control system
c) A drawing system
d) A sound system

8. Which Scratch block is used for the decision-making part of a control system?

a) 'Say Hello'
b) 'Go to X Y'
c) 'Clear graphic'
d) 'If/Then' block

Answers:

1.b	2.a	3.b	4.a	5.c	6.d	7.b	8.d

Lesson 22: Controlling Systems Remotely

1. What is a term for remotely changing a system's operation?

a) Remote control
b) Manual control
c) Local control
d) Direct contact

2. What is the purpose of a Central Heating System?

a) To cool a room
b) To clean the air
c) To distribute heat from a central source
d) To manage data backups

3. In what way do current remote controllers often connect to the system they operate?

a) Via fiber optic cable
b) Through coded light pulses (IR)
c) Using ultrasonic sound waves
d) By direct electrical current

4. Which device usually provides the input for an Air Conditioning control system?

a) A monitor
b) A printer
c) A speaker
d) A temperature sensor

5. Why is remote control especially useful for large industrial systems?

a) It allows safe operation
b) It checks equipment color
c) Tracks material count
d) It Creates loud alarms

6. Which physical system is controlled by an Air Conditioning unit?

a) Water flow pressure
b) Room temperature and cooling
c) Electrical power usage
d) Mechanical movement

7. What is the technical term for the temperature you set on a thermostat?

a) The variable input
b) The sensor reading
c) The set point
d) The final output

Answers:

1.a	2.c	3.b	4.d	5.a	6.b	7.c

Lesson 23: Climate Control Technologies

1. What is the main goal of Climate Control technologies?

a) Adjusting temperature and humidity
b) Measuring room pressure
c) Recording building dimensions
d) Monitoring electrical current

2. How do climate control technologies work?

a) Waste electricity
b) Create noise
c) Adjust air, heat, and moisture
d) Cause discomfort

3. Which devices are commonly used in greenhouse climate control?

a) Smartphones
b) Sensors and fans
c) Printers
d) Televisions

4. Where else are climate control systems commonly used?

a) Kitchen
b) Bedroom
c) Garage
d) Stadium/Airport

5. Which of the following is a common greenhouse heating method?

a) Solar heaters
b) Refrigerators
c) Microwaves
d) Electric ovens

6. Why is climate control technology important for health?

a) To keep indoor air safe and clean
b) To ensure proper air quality
c) To regulate airflow in buildings
d) d) To prevent Mold and dampness

7. What is the role of cooling systems in climate control?

a) Provide light
b) Store equipment
c) Water plants
d) Reduce high temperatures

8. How do climate control technologies contribute to energy efficiency?

a) By making everything extremely hot
b) By consuming excessive energy
c) By saving energy, leading to cost savings and environmental benefits
d) By creating a tropical paradise

Answers:

1.a	2.c	3.b	4.d	5.a	6.b	7.d	8.c

Lesson 24: Drone Technology

1. What does the acronym UAV stand for?

a) Utility All Vehicle
b) Underground Auto Van
c) Unmanned Aerial Vehicle
d) Universal Air Velocity

2. What do you understand by drone?

a) Unmanned aircraft
b) Toy Plane
c) Remote car
d) Small boat

3. What technology helps a drone know its location?

a) Wi-Fi router
b) GPS
c) Loud speakers
d) Printer cable

4. What is one of the diverse applications of drone technology mentioned?

a) Surveillance
b) Aerial photography
c) Package delivery
d) Environmental monitoring

5. Why are drones popular for photography?

a) Fly without power
b) Cheaper than cameras
c) Reduce soil use
d) Can reach high or difficult angles

6. What is the main use of drones in delivery?

a) Survey land or traffic
b) Deliver packages remotely
c) Assist in emergency rescue
d) Deliver mail

7. What is one use of drones in agriculture?

a) Water system Monitoring crops from above
b) Spraying pesticides or fertilizers
c) Mapping farmland
d) Assessing irrigation needs

8. What is the main benefit of using drones for inaccessible areas?

a) They make noise
b) They allow safe inspection
c) They fly fast
d) They are cheap

Answers:

1.c	2.a	3.b	4.a	5.d	6.b	7.a	8.b

Lesson 25: Programming a Robot Lawnmower

1. What does the term "automated" mean?

a) Controlled by computer
b) Operated manually
c) Controlled by nature
d) Operated using a remote

2. What does the output of the lawnmower system do?

a) Measures grass
b) Checks battery
c) Maps the garden
d) Cuts the grass

3. What sensor does the lawnmower use to avoid obstacles?

a) Light sensor
b) Contact sensor
c) Temperature sensor
d) Loudness sensor

4. In the Scratch program, what sprite represents the lawnmower?

a) The robot sprite
b) The grass
c) The flower
d) Print database

5. What kind of system lets a machine operate on its own without a human pilot?

a) Manual
b) Wireless
c) Automated
d) Wired

6. Which block determines the angle the lawnmower turns after hitting an obstacle?

a) 'Turn degrees'

b) Go to X Y'

c) 'Repeat until'

d) 'Set size to'

7. Which block helps a robot lawnmower sense grass edges and objects?

a) Looks Block

b) Motion Block

c) Sound Block

d) Sensing Block

8. For the lawnmower to operate continuously, which type of Scratch block must contain the movement code?

b) A move block

c) A 'Forever' loop

a) A 'Wait' block

b) A repeat block

Answers:

1.c	2.d	3.b	4.a	5.c	6.a	7.d	8.b

Lesson 26: Programming a Simple Traffic Control System

1. What is the main goal of a traffic control system?

a) To calculate pedestrian/walker count
b) To manage vehicle flow
c) Track road conditions
d) To Reduce parking violations

2. What color is the traffic light that tells cars to stop?

a) Yellow
b) Green
c) Red
d) Blue

3. Which sprites are important in a traffic simulation?

a) Cat, car
b) Crossing, light
c) Light, car
d) Cat, road

4. What type of system does a traffic signal represent?

a) A manual system
b) A timed control system
c) A simple feedback system
d) A hydraulic power system

5. What benefits does a traffic control system bring to a community or city?

a) Increasing accidents
b) Causing traffic jams
c) Creating confusion
d) Enhancing safety and traffic flow

6. What Scratch block is used by the traffic light to change its color?

a) 'Switch costume to'
b) 'Move steps'
c) 'Set size to'
d) 'Change X by'

7. What makes the cat look like it's walking in Scratch?

a) Playing a noise sound
b) Using single costume
c) Rapidly switching two costumes
d) Changing the background image

8. What is the process of finding and fixing mistakes in computer code?

a) Compiling
b) Debugging
c) Testing
d) Executing

Answers:

1.a	2.c	3.c	4.b	5.d	6.a	7.c	8.b

Lesson 27: Assistive Technologies

1. What is the main purpose of Assistive Technology?

a) To speed up internet connection
b) To help people with tasks they find difficult
c) To protect computers from viruses
d) To manage large factory equipment

2. Which group benefits most from assistive technologies?

a) Athletes
b) Programmers
c) Teachers
d) Disabled Person

3. What is an example of assistive technology for someone with low vision?

a) Fast computer
b) Loud speakers
c) Screen reader software
d) Small books

4. How are assistive technologies helping people in cars?

a) By doubling the car's fuel efficiency
b) By improving safety
c) By automatically finding the nearest parking spot
d) By adjusting the car's suspension height

5. What is the main benefit of using computer control in assistive car systems?

a) To allow precise, fast reaction
b) To make the wheels turn
c) To turn off the lights
d) To check the engine

6. What technology in cars keeps a car in the middle of its lane?

a) Blind spot monitor

b) Cruise control

c) Parking assist

d) Lane keep assist

7. What is the purpose of Screen Reader software?

a) To make text smaller

b) To read displayed text aloud

c) To highlight text on screen

d) To assist people with visual impairments

Answers:

1.b	2.d	3.c	4.b	5.a	6.d	7.b

Lesson 28: Programming a Game of Chance

1. What is a key feature of a Game of Chance?

a) Requires long skill practice
b) Result depends on luck or unpredictable events
c) Decided by fast reflexes
d) Involves complex math

2. Where in Scratch do you rename a sprite's costumes?

a) The Code tab
b) The Sound tab
c) The Costumes tab
d) The Backdrop area

3. In a game of chance, what is the value that changes randomly for each spin?

a) The random variable
b) The fixed constant
c) The system constant
d) The controlled parameter

4. Which two concepts play essential roles in electronic games of chance?

a) Skill and strategy
b) Speed and predictability
c) Complexity and simplicity
d) Probability and randomness

5. What is the most difficult part of making an electronic game of chance fair?

a) Making the graphics look good
b) Ensuring the randomness is unbiased
c) Programming the rules correctly
d) Adding sound effects

6. Which operator checks if all random values are equal for a jackpot?

a) OR

b) LESS THAN

c) AND

d) NOT

7. What is the primary purpose of incorporating randomness in a game?

a) To add surprise and variety

b) To make sprite movements predictable

c) To make the game less engaging

d) To reduce the use of color in the game

Answers:

1.b	2.c	3.a	4.d	5.b	6.c	7.a

Lesson 29: Programming a Multi Control Traffic System

1. What is the main purpose of a remote-control program in Scratch?

a) To control the movement of a sprite
b) To change the appearance of the stage
c) To change the appearance of the sprite
d) To link with other Scratch users

2. What does the broadcast block do in Scratch?

a) Change the sprite's costume
b) Send a message to other sprites
c) Make the sprite bigger or smaller
d) Move the sprite

3. What is a system that manages two or more controlled elements at once called?

a) A simple system
b) A manual system
c) A Multi-Control System
d) A single loop

4. How does a Multi Control Traffic System enhance safety?

a) By increasing congestion
b) By reducing travel times
c) By causing unnecessary stops
d) By coordinating signals at intersections

5. What must the cars do when they receive the 'Stop' message from the traffic light?

a) Stop moving
b) Speed up
c) Turn around
d) Sound the horn

6. What is the main advantage of using a multi-control system in a city setting?

a) Creates more parking
b) Makes cars travel faster
c) Reduces road repair
d) Efficient traffic management

7. In Scratch, what does the concept of "layer" refer to?

a) Different programming languages
b) The order in which images are displayed on the screen
c) The number of roads in a traffic system
d) The type of vehicles used in the game

Answers:

1.a	2.b	3.c	4.d	5.a	6.d	7.b

Lesson 30: Programming a TV Remote Control

1. What is the definition of a TV Remote Control?

a) A device used to control a car
b) A handheld device for controlling a television from a distance
c) A garage door opener
d) A game controller

2. What kind of power source does a remote control require to deliver a signal?

a) Wall socket
b) USB cable
c) Internal battery
d) Solar energy

3. What technology does a typical TV remote control use to send signals?

a) Sprite Bluetooth waves
b) Infrared light
c) Sound waves
d) Radio antenna

4. What is the primary advantage of using an Infrared signal over a simple wire connection for a remote?

a) It allows wireless control
b) It is much cheaper
c) It uses more power
d) It is always faster

5. What is the purpose of the Infrared light in a remote-control system?

a) To light up the room
b) To charge the TV
c) To measure distance
d) Transmit the coded command

6. Which Scratch feature changes story events?

a) Switch backdrop
b) Forever loop
c) If-then block
d) Repeat block

7. What does "IR" stand for in the context of TV remote controls?

a) Internet Relay
b) Infrared
c) Intelligent Remote
d) Inverse Receiver

8. What does dual function mean?

e) Performs one task only
f) Works with one device
g) Performs multiple tasks
h) Limited to a single function

Answers:

1.b	2.c	3.b	4.a	5.d	6.c	7.b	8.c

Lesson 1: Introduction to Handling and Modelling Data

1. What is data?

a) A collection of facts
b) Numbers
c) Statistics
d) All of the above

2. What is the first step in data modeling?

a) To create data
b) To store data
c) To represent and analyze data using visual tools
d) To delete data

3. What is the primary function of a Data Capture Form?

a) Organize results
b) Collect raw data
c) Calculate averages
d) Practice digital drawing skills

4. What does data investigation involve?

a) Collect, organize, present
b) Write essays only
c) Paint charts
d) Read stories

5. Who is called a respondent in a data investigation?

a) Question maker
b) Response provider
c) Class teacher
d) Data table

6. Which software is best suited for creating a soft copy of the data capture form before distribution?

a) Spreadsheet software
b) Presentation software
c) Word processing software
d) Simulation software

7. How do graphs help when studying data?

a) They show patterns
b) They save money
c) They make things colorful
d) They reduce effort

8. What does "validity" mean?

a) How fast the data is collected
b) How correctly the data measures the right thing
c) How neatly the data is written
d) How many people filled the form

Answers:

1.d	2.c	3.b	4.a	5.b	6.c	7.a	8.b

Lesson 2: Capturing Data

1. What do you understand by data capture?

a) Removal of data
b) Creating graphs
c) Share data with others
d) Recording information correctly

2. Which method is commonly used to gather opinions from many students?

a) Attendance register
b) Survey form
c) Observation log
d) Classroom notes

3. Why should the team ensure every sample member has responded?

a) To practice counting
b) To keep records neat
c) To have complete data
d) To finish faster

4. What is an alternative method to collect data without using forms?

a) Observation
b) Survey forms
c) Questionnaires
d) Attendance register

5. In the Capturing Data step, what is the main thing to collect?

a) Final conclusions
b) Charts and graphs
c) Complete, accurate raw data
d) Presentation notes

6. What is the container/counter method used to study?

a) Cost of materials
b) Efficiency and accuracy of data collection
c) Best font to use
d) Form color

7. How does the counter method help with discussion and comparison?

a) Makes patterns easier to see
b) Uses less technology
c) Speeds up understanding
d) Shows results visually

8. Why should data be accurate?

a) Reliable results
b) Save paper
c) Add color
d) Entertainment

Answers:

1.d	2.b	3.c	4.a	5.c	6.b	7.d	8.a

Lesson 3: Organizing Data

1. Where is raw data first put?

a) Storyboard page
b) Data table
c) Drawing chart
d) Color sheet

2. Which software helps to organize and handle numerical data easily?

a) Spreadsheet program
b) Paint software
c) Text editor
d) Drawing tool

3. What do we use to show patterns or trends in collected data?

a) Video clips
b) Photo slides
c) Story maps
d) Charts or Graphs

4. Which type of chart can be used to represent data in a circular format?

a) Line chart
b) Bar chart
c) Pie chart
d) Flow chart

5. Why do we use headings in a spreadsheet?

a) To format columns
b) To identify data
c) To sort information
d) To filter records

6. Why is it important to tally raw data before entering it into a spreadsheet?

a) To ensure all responses are counted
b) To verify no responses are missing
c) To check for any counting errors
d) Prepare data for digital entry

7. Why to use line chart for time data?

a) Easier to draw
b) More colors
c) Show trends over time
d) Vertical lines

Answers:

1.b	2.a	3.d	4.c	5.b	6.a	7.c

Lesson 4: Using Spreadsheets

1. What do you understand by spreadsheet?

a) A type of chart
b) A tool to write a reflection
c) A tool for organizing and analyzing data
d) A word processing software

2. What essential feature must all charts and graphs include to be easily understood?

a) Footers
b) Clear headings
c) Formulas
d) The sheet number

3. What term refers to a page in a spreadsheet?

a) Document
b) Slide
c) Worksheet
d) File

4. What feature allows quick calculation?

a) Colors
b) Fonts
c) Pictures
d) Formulas

5. Which of the following is not a type of chart or graph suggested for use?

a) Pie chart
b) Bar chart
c) Line chart
d) Scatter plot

6. What term refers to visual representations showing relationships between data sets?

a) Spreadsheets
b) Graphs
c) Histograms
d) Tables Bars

7. What does review for consistency in charts mean?

a) All charts have same title
b) Numbers must be bold
c) Same color represents same item
d) Sort all charts descending

8. Which menu is used to insert chart & tables in a spreadsheet?

a) Home
b) Design
c) Draw
d) Insert

Answers:

1.c	2.b	3.c	4.d	5.a	6.b	7.c	8.d

Lesson 5: Preparing a Data Presentation

1. Why is it important not to add too much text on slides?

a) Reduces clarity
b) Increases interest
c) Enhances visibility
d) Improves accuracy

2. Which element improves audience engagement in a presentation?

a) Long notes
b) Paragraph text
c) Number lists
d) Data charts

3. What should the team aim to include on the slides besides just text?

a) Complex formulas
b) Supporting charts and pictures
c) Raw data forms
d) Spreadsheet file

4. What can graph, charts, and images contribute to data presentation?

a) Make slides longer
b) Add extra numbers
c) Enhance understanding and engagement
d) Fill empty space

5. What is the key design principle for a good presentation?

a) Only black background
b) Keep it clear and readable
c) Many complex visuals
d) Make it very long

6. What should you do with complex information in a data presentation?

a) Add extra charts
b) Use long paragraphs
c) Break it into bullet points
d) Combine into fewer slides

7. What should you avoid when presenting data effectively?

a) Cluttered design
b) Apply appropriate colors
c) Give short explanations
d) Use readable fonts

Answers:

1.a	2.d	3.b	4.c	5.b	6.c	7.a

Lesson 6: Delivering a Data Presentation

1. How does practicing your presentation help?

a) Helps you choose colors for slides
b) Builds confidence and smooth delivery
c) Let's you collect new data
d) Makes charts automatically

2. What skill is demonstrated by clearly presenting to an audience and engaging it?

a) Formatting skill
b) Communication skill
c) Spreadsheet skill
d) Data entry skill

3. What type of feedback are students encouraged to give to their peers?

a) Only positive comments
b) Only negative criticism
c) Constructive suggestions for improvement
d) No feedback at all

4. What is an important role of the audience during presentations?

a) Active listening
b) Question writing
c) Slide reading
d) Note decorating

5. After a presentation, what should be provided to the speaker?

a) Extra slides
b) Printed report
c) Constructive feedback
d) Audience notes

6. Which part of a slide should be limited?

a) Text
b) Title
c) Images
d) Colors

7. What is a good way to handle questions from the audience?

a) Answer quickly without thinking
b) Ask someone else to answer
c) Give a long unrelated explanation
d) Answer politely and clearly

8. What should a presenter do if they forget a point during the presentation?

a) Skip the rest of the presentation
b) Pause and continue calmly
c) Read the slide word by word
d) Rush to the next slide

Answers:

1.d	2.b	3.c	4.a	5.c	6.a	7.d	8.b

Lesson 7: Reflecting On and Improving Your Digital Work

1. What is the main purpose of reflection in digital projects?

a) To review strengths and weaknesses
b) To improve project ideas
c) To check workflow steps
d) To evaluate design choice

2. Which of the following can be set as a future target after reflection?

a) Use fewer colors
b) Forget digital techniques
c) Increase coding speed
d) Improve presentation skills

3. What outcome should reflection include?

a) Positive points about work
b) Random notes
c) Deleted slides
d) Pictures

4. Which part of your digital work can you improve in formatting?

a) The length of the data items
b) Chart colors and labels
c) The physical quality of the paper
d) The total count of respondents

5. What is the final stage to finish the investigation project?

a) Designing a new investigation
b) Sending out the data forms
c) Completing the reflection
d) Drawing the final chart

6. What should you set for yourself after answering the reflection questions?

a) A reward
b) One target for improvement
c) A positive statement
d) A grade

7. How does reflecting on teamwork skills improve digital work?

a) Shows who did more work
b) Only helps your partner
c) Doesn't improve digital work
d) Shows supportive skills for future teamwork

Answers:

1.a	2.d	3.a	4.b	5.c	6.b	7.d

Lesson 8: Simulating Real-World Environments

1. Which profession makes frequent use of flight simulators?

a) Aviation pilots
b) Graphic designers
c) Historians
d) Librarians

2. What do simulations provide that real experiments sometimes cannot?

a) Limited testing scope
b) Expensive trial costs
c) Physical injuries
d) Safe practice environment

3. Which type of simulation is commonly used in medical training?

a) Patient monitoring
b) Surgery practice
c) Data entry systems
d) Health record design

4. Which field often uses ship simulators?

a) Music Industry
b) Graphic design
c) Marine training
d) Food science

5. Which arrow keys were used to control the helicopter in the suggested simulation game?

a) WASD keys
b) Mouse only
c) Space bar
d) Up, Down, Left, Right arrows

6. What is the role of complex mathematical algorithms in simulations?

a) To calculate player scores
b) To solve simple equations
c) To simulate interactions and behaviors realistically
d) To create simple scenarios

7. Why are models useful for prediction and testing?

a) Test and observe without affecting reality
b) Uses simple code
c) Calculates averages
d) Only for entertainment

8. Which skills improve in timed or stressful simulations?

e) Data entry
f) Decision-making and reaction
g) Presentation
h) Formatting

Answers:

1.a	2.d	3.b	4.c	5.d	6.c	7.a	8.b

Lesson 9: Different Simulation Technologies

1. IMAX and VR are examples of which technology?

a) Printed graphics
b) Digital paintings
c) Musical devices
d) Simulation systems

2. What special equipment might be needed for some high-end simulations?

a) A standard keyboard
b) A regular mouse
c) VR headsets
d) A normal monitor

3. Which feature primarily makes VR simulations feel more realistic for users?

a) Interactive controls
b) Sound effects
c) Haptic feedback
d) 3D immersion

4. Besides training, what is simulation increasingly used for?

a) Entertainment purposes
b) Data capture
c) Report writing
d) Spreadsheet creation

5. Which game often includes VR?

a) Card shuffling
b) Crossword puzzle
c) Arcade racing
d) Dice rolling

6. Why are hurricane or earthquake simulators sometimes featured in science museums?

a) To test equipment
b) To train engineers
c) To show off graphics
d) To let people safely experience the scenario

7. "In VR simulations, what does immersion mean??

a) Cost of the simulation
b) Feeling present in the virtual world
c) Computer speed
d) Setup time

Answers:

1.d	2.c	3.d	4.a	5.c	6.d	7.b

Lesson 10: Programming a Simulation Game

1. What skills can be developed by creating a simulation game'\?

a) Graphics designing skills
b) Problem-solving, critical thinking, programming
c) Animation and sound skills
d) Game testing and debugging

2. Why is realism important in emergency simulation games?

a) It improves decision-making
b) It increases storage space
c) It simplifies passwords
d) It reduces screen size

3. What feature makes a simulation game different from other games?

a) Character customization
b) Complex controls
c) Score tracking
d) Realistic mode

4. In a racing game, what is the purpose of the racing timer?

a) To change the background
b) To record the user's performance
c) To move the car automatically
d) To play music

5. How do simulation games encourage curiosity and learning?

a) By giving random tasks
b) By removing challenges
c) By providing hands-on experience
d) By showing different scenarios

6. In Scratch, what does a variable usually store in a number game?

a) User's guess and system's number
b) Sprite color
c) Stage background
d) Number of blocks used

7. What action includes assigning a variable its initial value?

a) Declaration
b) Iteration
c) Assignment
d) Initialization

Answers:

1.b	2.a	3.d	4.b	5.c	6.a	7.d

Lesson 11: Using Design Software

1. What is the primary focus of using design software in interior design?

a) Choosing furniture randomly
b) Practicing color mixing
c) Enhancing space planning and selection
d) Making spaces look chaotic

2. How would you define interior design?

a) Arrange furniture and decor
b) Paint building walls
c) Design gardens
d) Plan city roads

3. Which option allows resizing an object in design software?

a) Rotate tool
b) Scale tool
c) Crop tool
d) Erase tool

4. What advantage does 3D design software give to product makers?

a) Print layouts
b) View shapes
c) Rotate objects
d) Test models

5. Which program is used for room layout?

a) Photoshop
b) Home styler
c) WordPad
d) Audacity

6. Which feature helps arrange objects neatly?

a) Erase tool
b) Crop tool
c) Align tool
d) Sound tool

7. What is the primary benefit of using design simulation over traditional hand-drawn plans?

a) It only works on tablets
b) It is faster to draw
c) It uses less paper
d) Provides a realistic visualization

8. How does interior design simulation help reduce real-world project costs?

a) Prevents costly mistakes
b) Removes the need for labor
c) Makes furniture cheaper
d) Lowers software price

Answers:

1.c	2.a	3.b	4.d	5.b	6.c	7.d	8.a

Lesson 12: Technologies to Support Training

1. What is VR in training for emergency services?

a) Fire hose operation tool
b) Immersive training technology
c) First aid kit
d) Hazard response trainer

2. What do fire services also perform besides firefighting?

a) Rescue operations
b) Music shows
c) Sports events
d) Market sales

3. Which safety practice is commonly used in schools?

a) Earthquake drill
b) Sports training
c) Fire drill
d) Music rehearsal

4. Why are simulations useful in emergency training?

a) Entertainment
b) Cost increase
c) Public display
d) Risk reduction

5. Besides environment, scenarios, and skills, what should a storyboard define?

a) Colors
b) Names
c) Technology used
d) Dates

6. What is the main benefit of watching a recorded simulation training session?

a) Entertainment
b) Helps review and improve performance
c) Reduces cost
d) Speeds up coding

7. Why is repeating a scenario useful in emergency training?

a) Keeps a record
b) Makes training longer
c) Improve response skills
d) Let's trainees practice rare or critical situations

Answers:

1.b	2.a	3.c	4.d	5.c	6.b	7.d

Lesson 13: Technologies to Improve Safety

1. Which tool is often used in vehicle crash tests?

a) Crash dummies
b) Traffic lights
c) Car tires
d) Seat covers

2. Why do car makers use simulations?

a) Build faster cars
b) Reduce accident impact
c) Sell more cars
d) Design luxury cars

3. What technology records results during crash testing?

a) Video cameras
b) Traffic signals
c) GPS trackers
d) Radio sets

4. Which variable is commonly tested in crash simulations?

a) Fuel Level
b) Braking force
c) Engine size
d) Vehicle speed

5. Which system improves safety by reacting automatically in emergencies?

a) Anti-Lock brakes
b) Airbag system
c) Traction control
d) Lane assist

6. What do crash test simulations help to reduce?

a) Injuries and deaths
b) Car colors
c) Petrol prices
d) Road traffic

7. Which group benefits most from safer car design?

a) Mechanics
b) Car companies
c) Car painters
d) Drivers and passengers

Answers:

1.a	2.b	3.a	4.d	5.b	6.a	7.d

Lesson 14: Testing Aerodynamics

1. What does aerodynamics mainly study?

a) Air movement
b) Water flow
c) Sound waves
d) Heat energy

2. What is the main advantage of virtual wind tunnels?

a) Physical testing
b) Faster cars
c) Virtual airflow
d) Ride games

3. Which force do engineers evaluate in aerodynamics?

a) Gravity force
b) Magnetic force
c) Tension force
d) Drag force

4. Why do engineers reduce drag in car design?

a) Slow cars
b) Fuel saving
c) Air block
d) Heavy cars

5. What is the process of checking if a simulation accurately models the real world called?

a) Tallying results
b) Data collection
c) Validation and Verification
d) Reviewing the model

6. How do simulations help design wind turbines?

a) Create fake electricity
b) Find the best turbine shapes
c) Build fun virtual rides
d) Find the best turbine shapes

7. Which of the following is NOT an application of aerodynamics simulations?

a) Cooking food
b) Changing airplane shapes
c) Making cars use less fuel
d) Planning roller coasters

8. Why is a well-designed simulation useful for predictive modeling?

a) Saves experiment time
b) Predicts system behavior
c) Keeps data organized
d) Shows results clearly

Answers:

1.a	2.c	3.d	4.b	5.c	6.d	7.a	8.b

Lesson 15: Testing Design Structure

1. Why are simulations preferred over constant prototypes?

a) More efficient
b) Less accurate
c) Too complex
d) Very slow

2. What is the definition of design structures?

a) Materials used in construction
b) Plans and frameworks
c) Tools for building
d) Workers on site

3. How do simulations benefit engineers and architects in testing design structures?

a) Reduce material waste
b) Avoid early prototypes
c) Test safety virtually
d) Estimate building time

4. What is the primary focus of testing design structures?

a) Safety & functionality
b) Cost-cutting
c) Luxurious buildings
d) Estimating workers

5. How do simulations help in making buildings more eco-friendly?

a) Compare airflow and light
b) Try different building shapes
c) Check basic material choices
d) Test energy-saving designs

6. In which area of design structures is skyscraper design important?

a) Furniture design
b) Landscape design
c) Designing tall buildings
d) Interior design

7. What concept does simulation test before final construction?

a) Material cost prediction
b) Feasibility under stress
c) Worker schedule check
d) Presentation slide design

8. Why should we test Design Structures?

a) To check if the structure looks neat
b) To see if the structure follows the design
c) To check if the structure is easy to build
d) To ensure the structure is safe and strong

Answers:

1.a	2.b	3.c	4.a	5.d	6.c	7.b	8.d